

NATURAL AND ANTHROPOGENIC INFLUENCES RECORDED IN SEDIMENTS FROM THE KRKA RIVER ESTUARY (EASTERN ADRIATIC COAST), EVALUATED BY STATISTICAL METHODS

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ABSTRACT

Surface sediments and cores from the Krka river estuary (Eastern Adriatic coast) were analyzed for their trace-element composition by means of HR ICP-MS (High Resolution Inductively Coupled Plasma Mass Spectrometry). Various statistical analyses were used to evaluate the geochemical composition of these sediments, and to establish the degree of anthropogenic influence to the Krka river estuary.

Statistical analysis showed clear anthropogenic influence on several spots of the Krka river estuary. Sediments sampled in the Šibenik port demonstrated the highest concentrations of eco-toxic metals as being Cd (34.4 mg/kg), Zn (4010 mg/kg), Pb (253 mg/kg), As (45.3 mg/kg) and Cu (24.6 mg/kg). Calculated Müller's geoaccumulation index (Igeo) values, after normalization on average carbonate values, are for Cd (4.7), Zn (6.9), Pb (4.2), As (5.5), and Cu (2.0). Igeo values, after normalization on average shale, are for Cd (6.3), Zn (4.9), Pb (3.1), As (1.6), and Cu (1.6), suggesting a moderately to heavily polluted Šibenik harbor. The pollution source at this location results from long-term phosphate ore unloaded at the port, as well as input of untreated wastewaters from the city of Šibenik. Another contaminated area was established in front of the now closed ferromanganese factory, with maximum levels for Mn (2526 mg/kg), Pb (143 mg/kg) and Ba (255 mg/kg). However, in the major portions of the Krka river estuary, concentrations of measured elements were low and varied only due to natural (lithogenic and marine) influences. This indicates that the majority of the Krka river estuary is still unpolluted.

KEYWORDS: Krka river estuary, Eastern Adriatic coast, sediment, metals, anthropogenic influence, statistical analysis.

INTRODUCTION

Estuarine areas are complex and dynamic aquatic environments, which are often suitable for the development of urban and industrial centers, and are thus prone to anthropogenic influences. For many years, industrial, agricultural and domestic wastes were discharged into waterbodies without sufficient control, or understanding of their pollution effects. Pollutants in those wastes, particularly toxic metals, can endanger public health by entering the food chain [1]. Metals are not biologically degradable and they tend to accumulate, especially in aquatic sediments associated with organic and inorganic matter. It is important to distinguish between the introduction of trace metals by anthropogenic activities or natural weathering processes.

The sediments are not only the reservoir for contaminants, but also a possible source of toxicants for aquatic organisms, if remobilized due to physio-chemical changes in the water column [2]. Therefore, monitoring of trace metals in sediments is necessary to provide insight into recent, as well as historic anthropogenic pollution. Natural variations in trace elements' content in aquatic sediments can make recognition of the anthropogenic influence difficult, especially in estuarine environments where both terrigenous and marine influences affect the geochemical composition of sediments.

In order to recognize anthropogenic-derived deviations from the natural geochemical composition, multi-elemental analyses (Mass spectrometry) of sediment samples from this estuary and three different statistical methods were applied.

The study was performed on a set of data from the Krka river estuary (Eastern Adriatic coast). This is still a relatively unpolluted estuary [3-8], where some local contamination with Mn [9, 10], Hg [11, 12], U [13] and Ra [14] was recently established.

MATERIALS AND METHODS

Study area

The Krka River estuary was formed during the Holocene transgression. Located between Skradinski Buk wa-

terfalls (calc-tufa barrier) through the Prokljan Lake to the St. Nikola fortress, the estuary has a total length of 22 km (Fig. 1). The hydrogeological drainage area of the Krka River is approximately 2427 km². The estuary bottom gradually deepens from 2 m, below the waterfalls, to 42 m, in front of Fort St. Nikola. It is a typical karstic highly stratified estuary with fresh/brackish surface layer flowing seawards and bottom seawater counter-currently moving upwards. Input of terrigenous clastic material into the Krka river estuary is small, with the main input of the particulate material *via* the small Guduča River (Fig. 1) inflowing into Prokljan Lake [15, 16]. The Krka River carries larger quantities of fresh-water (on average 55 m³/s) than the Guduča River (on average less than 1 m³/s). However, a number of calc tufa barriers along the Krka River, upstream of the town of Skradin, significantly reduce suspended material transport [15-17].

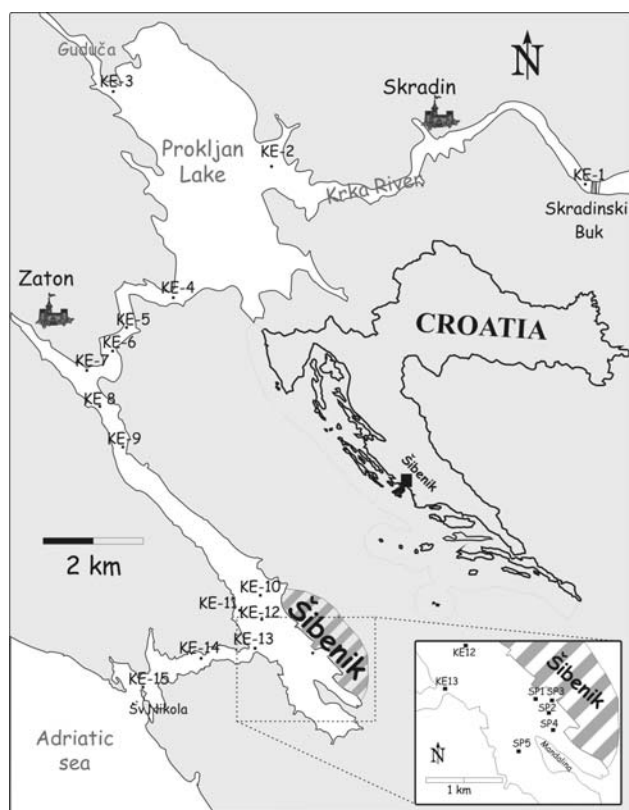


FIGURE 1 - Krka river estuary study area with sampling locations.

The largest settlement in the region is the town of Šibenik (~50 000 inhabitants), located in lower part of the Krka river estuary. Untreated city wastewater, old port facility and ferromanganese industry were main sources of contamination. Today, ferromanganese factory is closed, port facilities are reconstructed, and city wastewater is treated and taken out of the estuary.

Sediment input in the Krka River estuary from Krka River is relative small because of those tufa barriers which effectively catch most of sediment [19]. Sediment rate in Krka River estuary varies from 4-5 mm annually, at the

enter of Guduča River to Prokljan Lake, to less than 1 mm per year, in the region of Šibenik city [20].

Sampling and sampling preparation

Sediments were collected during 2004 and 2005, by a scuba diver using hand-driven acrylic corers (ϕ 14 cm), at 15 locations (KE-1 to KE-15) covering the whole longitudinal transect of the estuary (Fig. 1), starting from just below the Skradinski Buk waterfalls (KE-1) to the mouth of the estuary (KE-15). Additionally, sediments at 5 locations in the area of the Šibenik port (SP-1 to SP-5) were sampled by a scuba diver. Precise locations of sampling stations were determined using a GPS instrument (Garmin GPSMap 76 CSx, Kansas City, USA) with an accuracy of $\pm$ 5 m. At 12 locations, in the Krka estuary and in the Šibenik port, only surface sediments (first 5 cm) were sampled, but at 3 locations (KE-7; KE-10 and KE-13), sediment cores were also taken and sliced into 5 cm layers.

Sediments were wet sieved with ambient water using 0.063 mm standard Retsch sieves (Haan, Germany). Using ambient water and fast sieving, we diminish possibility of dissolving elements from fractions. Fine fraction ($<63 \mu\text{m}$) was separated for future analysis because it is known that fine fraction binds more pollution [4], therefore being recommended for evaluation of the sediment quality in terms of pollution [21].

After sieving, sediments were dried in laminar at room temperature until constant mass.

Sample digestion and analysis

Sediment subsamples (0.1 g) were digested with 10 ml of nitric acid supra pur (Fluka, Japan), in a microwave digestion oven (Multiwave 3000, Anton Paar, Graz, Austria). Samples were prepared in batches of 6, with one standard (MESS-3) and one blank. Quality control was provided by the preparation and measurement of certified reference marine sediment (MESS-3, NRC, Canada; see first two lines in Table 1). Digestion with nitric acid alone does not result in total digestion, but analysis of certified reference marine sediment (MESS-3, NRCC, Canada) revealed that, for all analyzed elements (except for Ti) almost quantitative recovery (80-100 %) was obtained. After digestion, samples were diluted with Milli-Q water and internal standard (In) was added.

Multi-elemental analysis of samples was performed by means of high resolution inductively coupled plasma mass spectrometry (HR-ICPMS) using the instrument Element 2 (Thermo, Bremen, Germany), characterized by 3 different resolutions: low resolution (LR), medium resolution (MR) and high resolution (HR). The following isotopes were measured at chosen resolutions: LR: ⁷Li, ⁹Be, ⁹⁵Mo, ⁹⁸Mo, ¹¹¹Cd, ¹¹⁴Cd, ²⁰⁵Tl, ²⁰⁸Pb, ²⁰⁹Bi; MR: ⁴⁷Ti, ⁵¹V, ⁵²Cr, ⁵⁵Mn, ⁵⁹Co, ⁶⁰Ni, ⁶³Cu, ⁶⁶Zn, ⁸⁶Sr, ¹³⁸Ba; HR: ²³Na, ²⁷Al, ⁵⁶Fe, and ⁷⁵As. For quantification of elemental concentrations, external calibration was used. Standards were prepared by appropriate dilution of a multi-elemental standard (Analytika, Czech Republic).

Statistical analysis

Two different statistical approaches, to identify anomalous geochemical data, were used as described by Frančišković-Bilinski [22]. In addition, factor analysis was performed.

The first statistical method was the box-plot method where normal or lognormal box-plots are constructed on the base of the empirical cumulative distribution plots. The box length was the interquartile range, where outlier values were defined as 1.5 and 3 box lengths from the upper or lower edge of the box. Outliers (extremes) were values more than 3 box lengths from the edge of the box [23, 24].

The second statistical method was the median absolute deviation (MAD). It is defined as the median of the absolute deviations from the median of all data [25]. The median value ± 2 MAD defines a fence, which separates outliers and extremes from a population.

The third method was the factor analysis. This method differs from PCA, as it is usually considered as a statistical technique. From "regular observations" the PCA method extracts the main elemental associations, which control the multivariate assignment of database. In factor analysis, the relation between a set of m variables is assumed to reflect correlations of every one of the variables with p mutually non-correlated main factors. The general assumption is that $p < m$. Variance of m variables is derived from the variance of the p factor. More theoretic details can be found in Davis [25] and Halamić et al. [26].

RESULTS

Results of multi-elemental analysis geochemical composition of Krka river estuary sediments are presented in Table 1. We performed a statistical evaluation on two datasets. The first dataset consisted of all 26 analyzed samples, including 5 samples from the Šibenik port (SP), for which we suspected a significant anthropogenic influence. The second dataset (21 samples) included only samples from the Krka river estuary (KE samples, Fig. 1) which has limited influence from pollution sources [3, 5, 7, 17]. The second statistical analysis was aimed to facilitate the recognition of possible contamination and anthropogenic influence in the clean part of the estuary. The results of the box-plot and MAD statistical methods of the first (extended) dataset are presented in Table 2, and of the second one in Table 3.

In the first dataset (Table 2), which includes the samples from the Šibenik port, according to the box-plot approach, extreme concentrations of eco-toxic metals were found only in samples from the Šibenik port. In all 5 samples, some elements (Cd, Pb, Zn, As) were elevated. Outliers were also obtained for other elements (Cu, As, Mo, Pb, Cr, Ba). From the residue of the Krka river estuary, only one location (KE-7 (5-10)) demonstrated outliers for Li, Be, Mo, Ti, V, Co, Al, Fe and As. The anomaly detection by the MAD method (Table 2) applied on the first dataset, pointed to the same locations (see box-plot method) as the polluted ones (5 locations in the Šibenik port), but with a larger number of elements with anomalous concen-

TABLE 1 - Element concentrations in sediments of the Krka river estuary (mg/kg).

Locations	Li	Be	Mo	Cd	Tl	Pb	Bi	Ti	V	Cr	Mn	Co	Ni	Cu	Zn	Sr	Ba	Na	Al	Fe	As
MESS-3 measured	73.6	2.3	2.78	0.24	0.9	21.1	-	4400	243	105	324	14.4	46.9	33.9	159	129	-	16000	85900	43400	31.2
	66.7	2.13	2.29	0.24	0.77	20.9	0.3	1950	230	95.9	295	13.5	46.9	31.4	148	117	691	13700	70300	37200	18.6
KE-1	16.1	0.49	0.51	0.44	0.49	13.1	0.13	218	22.1	20.5	124	1.96	14.8	9.0	24	418	40	2004	10488	5456	2.2
KE-2	64.7	1.73	1.47	0.57	1.00	35.0	0.38	503	83.9	76.4	396	7.83	47.7	18.6	111	520	131	10654	39571	21932	10.5
KE-3	52.1	1.88	1.12	0.54	0.98	26.6	0.36	566	99.2	93.4	430	9.72	62.9	24.2	70	346	108	12872	43797	26320	14.6
KE-4	74.2	2.04	1.18	0.58	1.13	60.6	0.78	637	93.0	93.0	974	10.8	65.1	26.0	117	901	126	11471	44885	26759	11.0
KE-5	49.2	1.47	0.76	0.44	0.82	60.4	0.80	502	66.6	70.4	1239	8.00	48.2	22.1	109	940	113	7620	34450	19388	7.2
KE-6	50.2	1.48	1.10	0.50	0.82	53.7	0.80	477	65.7	69.6	1025	7.96	47.3	21.6	115	849	113	9371	34652	19605	8.6
KE-7(0-5)	52.0	1.64	1.38	0.49	0.87	75.5	0.93	554	76.2	78.2	1061	9.27	51.2	29.1	145	704	146	9397	38421	22653	10.8
KE-7(5-10)	90.3	2.62	3.46	0.74	1.02	108	1.24	884	134	131	1155	15.4	81.7	49.5	181	993	225	12734	65473	39109	26.3
KE8	42.4	1.31	1.09	0.38	0.71	61.5	0.71	421	60.4	61.9	1038	7.09	41.3	19.8	147	1172	115	11378	29635	17734	8.7
KE-9	32.4	0.96	0.73	0.38	0.56	67.1	0.82	353	47.1	57.1	1222	5.93	34.2	18.2	124	1632	122	17763	23207	13372	6.7
KE-10(0-5)	39.2	1.26	0.98	0.52	0.62	95.4	1.98	454	57.5	75.0	2151	8.30	47.8	40.5	194	972	194	9018	26025	18652	8.9
KE-10(5-10)	35.1	1.05	1.87	0.49	0.09	93.4	2.09	378	52.2	68.9	1824	7.53	41.1	36.2	158	856	166	8166	26582	15771	9.1
KE-10(10-15)	49.9	1.55	1.85	0.73	0.25	133	3.05	506	73.0	93.2	2526	10.2	55.9	50.2	221	1135	243	10272	36126	22355	14.1
KE-10(15-20)	55.4	1.60	2.00	0.67	0.38	143	2.87	562	80.5	98.4	2147	10.8	61.4	53.9	226	1159	255	10082	39999	24802	17.5
KE-11	26.1	0.86	0.76	0.64	0.50	81.4	1.28	330	45.4	87.1	1525	6.12	33.0	26.8	116	1465	209	16556	19751	12553	8.5
KE-12	35.3	1.16	1.02	0.47	0.56	81.1	1.71	405	52.6	67.2	1534	7.55	44.3	34.3	184	887	153	9477	25824	17021	10.3
KE-13(0-5)	38.2	1.24	0.98	0.48	0.62	82.4	1.62	429	60.0	71.9	1473	8.24	50.8	31.4	133	1191	133	12223	27177	17319	6.7
KE-13(5-10)	49.3	1.37	1.25	0.53	0.22	79.6	1.77	457	72.1	88.7	1001	8.77	57.9	31.4	142	1394	149	11937	21520	19453	7.3
KE-13(10-15)	45.3	1.25	1.10	0.38	0.13	62.3	1.05	436	64.4	80.8	666	7.73	59.9	24.3	96	1413	118	11560	22648	17027	5.2
KE-14	29.2	0.94	0.64	0.38	0.46	63.6	1.16	371	49.0	63.6	931	6.52	41.5	31.4	105	1355	107	10046	21396	14087	6.8
KE-15	22.4	0.75	0.64	0.31	0.36	46.6	0.67	321	38.8	52.3	543	5.02	32.3	16.8	67	1464	87	10618	18022	10828	6.9
SP-1	16.3	1.36	2.77	24.6	0.51	253	3.02	517	86.8	112	758	6.29	29.1	125	3299	1495	230	9205	42263	11967	36.6
SP-2	11.0	0.64	1.28	4.03	0.25	90.3	1.49	402	41.0	43.0	379	3.19	17.3	32.3	503	1686	129	7353	16130	8333	13.4
SP-3	10.8	1.52	3.31	34.5	0.34	229	1.8	417	86.5	139	479	3.41	23.5	120	4010	1982	376	7387	39629	10025	45.2
SP-4	21.0	1.08	2.1	7.08	0.5	126	3.68	695	65.8	73.8	516	5.80	33.7	81.2	734	1153	195	9353	22405	14012	21.9
SP-5	36.0	1.24	1.79	1.65	0.55	172	2.74	991	67.2	71.3	1006	8.19	42.0	77.3	369	915	249	12534	25051	17426	15.8

TABLE 2 - Results of box-plot and MAD analyses of the complete dataset (26 samples).

Sample	Anomaly determined by box-plot		Anomaly determined by MAD
	Extreme	Outlier	
KE-1	-	-	
KE-2	-	-	Li, Ti
KE-3	-	-	Be, Ti, V, Na, Al, Fe
KE-4	-	-	Li, Be, Ti, Ti, Co, Al, Fe
KE-5	-	-	
KE-6	-	-	
KE-7(0-5)	-	-	
KE-7(5-10)	-	Li, Be, Mo, Ti, V, Co, Al, Fe, As	Li, Be, Mo, Ti, Ti, V, Cr, Co, Ni, Ba, Al, Fe, As
KE-8	-	-	
KE-9	-	Na	Na
KE-10(0-5)	-	-	Mn
KE-10(5-10)	-	-	
KE-10(10-15)	-	-	Bi, Mn, Ba
KE-10(15-20)	-	-	Pb, Bi, Mn, Ba
KE-11	-	Na	Na
KE-12	-	-	
KE-13(0-5)	-	-	
KE-13(5-10)	-	-	
KE-13(10-15)	-	-	
KE-14	-	-	
KE-15	-	-	
SP-1	Cd, Pb, Zn	Cu, As	Mo, Cd, Pb, Bi, Cr, Cu, Zn, Ba, Al, As
SP-2	Cd	Zn	Cd, Zn, Sr
SP-3	Cd, Zn, As	Mo, Pb, Cr, Cu, Ba	Mo, Cd, Pb, Cr, Cu, Zn, Sr, Ba, As
SP-4	Cd, Zn	-	Cd, Bi, Ti, Cu, Zn, As
SP-5	Cd	Ti	Cd, Pb, Bi, Ti, Cu, Zn, Ba

TABLE 3 - Results of box-plot and MAD analyses of the reduced dataset (21 samples excluding the Šibenik port samples).

Sample	Anomaly determined by box-plot		Anomaly determined by MAD
	Ex-treme	Outlier	
KE-1	-	-	-
KE-2	-	-	-
KE-3	-	-	Al, Fe, As
KE-4	-	-	Li, Be, Ti, Ti, Co, Ni, Al, Fe
KE-5	-	-	-
KE-6	-	-	-
KE-7(0-5)	-	-	-
KE-7(5-10)	Mo, As	Li, Be, Ti, V, Cr, Co, Al, Fe	Li, Be, Mo, Pb, Ti, Cr, Co, Ni, Cu, Zn, Ba, Al, Fe, As
KE-8	-	-	-
KE-9	-	Na	Sr, Na
KE-10(0-5)	-	-	Bi, Mn, Zn, Ba
KE-10(5-10)	-	-	Mo
KE-10(10-15)	-	Pb, Mn, Ba	Mo, Cd, Pb, Mn, Cu, Zn, Ba, As
KE-10(15-20)	-	Pb, Cu, Ba, As	Mo, Cd, Pb, Mn, Co, Cu, Zn, Ba, As
KE-11	-	Na	Sr, Ba, Na
KE-12	-	-	Zn
KE-13(0-5)	-	-	-
KE-13(5-10)	-	-	Sr
KE-13(10-15)	-	-	-
KE-14	-	-	-
KE-15	-	-	-

trations. The main difference between the box-plot and the MAD method was that in the residue of the estuary the MAD method showed anomalies for much more elements (Table 2) at several locations: KE-7(5-10), KE-2, KE-3, KE-4, KE-10(0-5), KE-10(10-15), KE-10(15-20).

In the second dataset, excluding the Šibenik port (Table 3), using the box-plot approach, only two elements (Mo and As) showed extreme values in sample KE-7(5-10). However, outliers for several elements (Pb, Mn, Ba, Cu, As) were detected in samples KE-7(5-10), KE-10(10-15) and

TABLE 4 - Results of factor analysis by elements on the reduced dataset (21 sample – excluding the Šibenik port samples).

Factor Loadings (Varimax normalized) (Krka.sta) Extraction: Principal components (Marked loadings are > .700000)			
Variable	Factor 1	Factor 2	Factor 3
Li	0.95975	0.069554	-0.017219
Be	0.97840	0.058702	-0.047000
Mo	0.74530	0.470100	-0.089782
Cd	0.61796	0.548244	-0.135707
Tl	0.66459	-0.489348	-0.169916
Pb	0.21244	0.946989	0.174366
Bi	-0.01585	0.978859	0.049557
Ti	0.97548	0.154997	0.001935
V	0.98835	0.076269	0.036502
Cr	0.83634	0.399489	0.275411
Mn	-0.00569	0.925994	0.072381
Co	0.90668	0.379043	0.096930
Ni	0.87916	0.254269	0.154803
Cu	0.40875	0.883698	0.004997
Zn	0.30074	0.879580	0.057160
Sr	-0.31266	0.282402	0.843072
Ba	0.38831	0.861147	0.149361
Na	0.24705	0.016833	0.894050
Al	0.97085	0.105034	-0.083991
Fe	0.97536	0.200772	0.004655
As	0.83771	0.390487	-0.014962
Explained variance	10.68844	6.459502	1.749497
Percent of total variance explained	0.50897	0.307595	0.083309

TABLE 5 - Results of factor analysis by samples on the reduced dataset (21 sample – excluding the Šibenik port samples).

Factor Scores (Krka.sta) Rotation: Varimax normalized Extraction: Principal components			
	Factor 1	Factor 2	Factor 3
KE-1	-1.68168	-1.20172	-2.56790
KE-2	0.80985	-1.10001	-0.73459
KE-3	1.26813	-1.38545	-0.29482
KE-4	1.32877	-0.82891	0.10349
KE-5	0.13637	-0.56339	-0.55859
KE-6	0.17909	-0.53900	-0.48299
KE-7(0-5)	0.52743	-0.22903	-0.67153
KE-7(5-10)	2.90799	0.41719	0.19661
KE-8	-0.21061	-0.51803	0.34882
KE-9	-0.74780	-0.53552	2.10288
KE-10(0-5)	-0.39580	1.10025	-0.56664
KE-10(5-10)	-0.73467	1.14528	-0.92815
KE-10(10-15)	0.05850	2.22253	-0.36817
KE-10(15-20)	0.42284	2.09402	-0.25942
KE-11	-0.77183	0.34901	1.54823
KE-12	-0.49252	0.59968	-0.57115
KE-13(0-5)	-0.30358	0.16182	0.48442
KE-13(5-10)	-0.11778	0.28251	0.90666
KE-13(10-15)	-0.28713	-0.43107	1.12326
KE-14	-0.78719	-0.21286	0.42850
KE-15	-1.10837	-0.82728	0.76110

KE-10(15-20). Anomaly detection by the MAD method in the second dataset shows the same locations as contaminated ones, but, in addition, anomalous values were found also at locations KE-4, KE-10(0-5) and KE-12.

More detailed statistical analysis was performed on the second dataset in order to recognize both natural and anthropogenic influences in unpolluted or less polluted parts of the Krka river estuary. Results of factor analyses are presented in Table 4 (factor loadings) and in Table 5 (factor scores for each sampling station). Three factors (varimax normalized) were extracted, and obtained results fulfill well with the stipulation of Morrison [27] that main components should explain at least 75 % of the total variance (in our dataset 89.99 % of the total variance is explained with three factors).

DISCUSSION

In the first dataset, all extreme values detected by the box-plot method were in sediments from the Šibenik port. The maximal obtained levels of eco-toxic metals were: Cd (34.4 mg/kg), Pb (253 mg/kg), and, for the others, Zn (4010 mg/kg), As (45.3 mg/kg) and Cu (24.6 mg/kg). The concentration of Cd was 50-100 times higher than at other locations in the estuary (Table 1), whereas the difference for the other toxic elements was smaller (2-10 times). One of the main sources of contamination in the Šibenik port was transport of phosphate ore which took place intensively in this port for about 20 years [13]. Recent investigation on this location established serious contamination of sediments and mussels with uranium and ascribed this to the significant loss of material to the sea bottom during the unloading of the ore [13]. It is known that phosphate ore may contain high concentrations of Cd and elevated levels of As, Zn and Cu [28-30]. Therefore, high levels of these metals in sediments of the Šibenik port can be ascribed to this pollution source. The source of elevated concentrations of other metals could be the untreated wastewaters of the city of Šibenik which were, until recently, introduced to the estuary at this location. Other transport activities in the port connected with industry of Šibenik (for example, the aluminum factory) could also be sources of metal pollution.

If we look to results of statistical evaluation by the box-plot method (Table 2), which is applied to the complete dataset, we can see that this method can recognize the anthropogenic influence only at the location of the Šibenik port, while all other stations seem to be unpolluted. The elevated level of several elements (mostly lithogenic), obtained in the deeper part (5-10 cm) of the sediment column at station KE-7, can be ascribed to irregular inputs of the terrigenous material from the flysch zone situated in the vicinity of this location. The MAD method detected elevated concentrations of the same lithogenic elements at more locations (in addition to KE-7, also at KE-2, KE-3 and KE-4), which is in good accordance with the well-known

input of terrigenous material to the estuary at locations KE-2 to KE-4 by the Guduča river [19]. Also the MAD method was more successful in recognizing the anthropogenic influence to the estuary, as anomalous concentrations of Bi, Mn and some other elements were detected at location KE-10, which is situated in front of the closed ferromanganese factory. At this location, slag material is deposited on the coast, resulting in washout of Mn and some other metals to the sea [8-10].

In the second dataset (excluding Šibenik port), the results of the box-plot method (Table 3) showed extreme values for Mo and As in the sample KE-7(5-10), and outlier values for several lithogenic elements. Thus, from both datasets it is evident that this sample contains higher concentrations of some elements that, as we discussed earlier, most probably originate from the nearby flysch zone. In this dataset, however, one other location (KE-10) showed outliers for Pb, Mn, Ba and As. This location is situated in front of the closed ferromanganese factory which has already been shown to be a contributor of Mn [8, 10] and Pb [31]. It is interesting that outliers are found only in deeper sediment layers suggesting that contamination with metals on this location is diminishing after the factory was closed. The MAD method points to the same locations (KE-7 and KE10) as the box-plot method, but the greater number of elements with elevated concentration is notable. However, three other locations are notable from the MAD method, KE-2 and KE-3 in the upper part of the estuary, and KE-12 in the central part of the estuary. First two locations showed anomalous values for typically lithogenic elements. These elevated concentrations reflect well the input of terrigenous material by the Guduča river, which is the main source of this material for the whole estuary [19]. Location KE-12 is situated in front of the city of Šibenik, and elevated Zn level in sediment is likely a consequence of long-term discharge of untreated municipal wastewater to this part of the estuary. These results indicate that MAD method is more suitable for analysis of geochemical data than the box-plot method (gives more anomalous values), which was already concluded in the study of Reimann et al. [24].

From the results of the factor analysis (Table 4), the origin of the elemental contributions to different estuarine sediments can be recognized. Analysis showed that 3 important factors can be isolated and assigned with a good geochemical explanation. Factor 1 showed excellent correlation (>0.95) with Li, Be, Ti, V, Al and Fe, and good correlation (0.75-0.88) with Mo, Cr, Co, Ni and As. This factor can be explained as "lithogenic", and it can be assumed that samples with higher values are influenced by terrigenous components eroded from the Eocene flysch terrains. Factor 2 showed excellent correlation (>0.92) with Pb, Bi and Mn, and very good correlation (0.86-0.88) with Cu, Zn and Ba. This factor can be explained as the "anthropogenic", and samples with higher values are under influence of local pollution sources (nearby abandoned factories, the Šibenik port and wastewaters of the city of Šibenik).

Factor 3 is in good correlation (>0.84) with Sr and Na. This factor can be explained as the “marine”, and samples with higher values are under higher influence of seawater and marine carbonate sedimentation.

Factor analysis done by sample location (Table 5) enables separation of interesting locations in the Krka river estuary. Samples with the strongest terrigenous influence (>2.9) were restricted to one layer (5-10 cm) at location KE-7, in front of the village of Zaton. The influence of the fly ash zone at this location was discussed earlier in the paper. The strong terrigenous influence (>1.2) is recorded also in sediment near the inflow of the Guduča river (KE-3) to the estuary, and also at the exit of the Prokljan Lake (KE-4). The strongest anthropogenic influence was recorded in sediment in front of the closed ferromanganese factory (KE-10). This influence is higher in the deeper layer (>2) than at the surface (>1.1), in accordance with the fact that this factory was closed 15 years ago. Marine influence is the most notable one at locations KE-9, KE-11 and KE-13. These influences in sediments are distributed similarly as reported before [19], and indicate the dominant carbonate marine sedimentation processes acting on the lower part of the estuary.

The present study shows clear anthropogenic influence recorded in the sediments of the Krka river estuary, in at least two areas. The first one is that of the Šibenik port, and the second is the location in front of the closed ferromanganese factory. It is important to estimate the degree of sediment contamination at these locations by comparison with environmental guidelines. In the area of the Šibenik port, which was the most polluted one, elevated concentrations of Cd, Zn, Pb, As and Cu were found. Using Sediment Quality Guidelines provided by NOAA and ANZECC & ARMCANZ [32, 33] we conclude that maximum concentrations of Cd, Zn and Pb are greater than ERM (toxicity effect range median). Maximum levels of As and Cu are between ERL (toxicity effect range low) and ERM. Thus, the area of the Šibenik port can be considered as being heavily polluted by toxic metals. The anomalies in Šibenik port can be expressed using the “geoaccumulation index” (Igeo), according to Müller [34]. This index was calculated as $I_{geo} = \log_2 C_n / 1.5 B_n$. Igeo describes the relationship between the measured elemental concentration (C_n) and the background concentration (B_n). Normalization of total concentrations can be performed using background values for elements either in shale or in carbonates, according to Turekian and Wedepohl [35]. Carbonate-normalized Igeo values for elements showing highest concentrations are Cd (4.7), Zn (6.9), Pb (4.2), As (5.5), and Cu (2.0). Shale-normalized Igeo values for those elements are Cd (6.3), Zn (4.9), Pb (3.1), As (1.6), and Cu (1.6). The sediments in the investigated region present a mixture of carbonate- and shale-like material in unknown proportions. Both kinds of normalization give Igeo values, according to Müller [34], and indicate moderately to highly polluted environment in Šibenik port. Fortunately, this contamination is especially localized, and spreading of the

polluted sediment from the area of the Šibenik port to the rest of the estuary is restricted by the bottom topography and the weak sea currents [13]. Sediments at the second polluted location, in front of the ferromanganese factory, showed elevated concentrations of Mn, Pb and Ba, but only in the deeper part of the sediment column (indicating reduction of pollution input in the last decade). Using the same sediment quality guidelines as above, we may conclude that concentrations of Pb are between ERL and ERM. The Technical Guidance for Screening Contaminated Sediments [36] was used as guideline for Mn, as in Sediment Quality Guidelines provided by NOAA and ANZECC & ARMCANZ [32, 33], values for Mn are not given. According to these guidelines, concentrations of Mn in sediments in front of the closed ferromanganese factory are above Severe Effect Level. So, this location can be also considered as polluted one. All other measured values for all elements at all locations are well below the notable effect level.

CONCLUSION

All three statistical methods employed in this study can be used to recognize the natural and anthropogenic influence from the geochemical composition of sediments. However, each of the method has specific advantages. By the box-plot method, the extreme values can be easily recognized. The MAD method gives similar results as the box-plot method, but it is more sensitive for recognizing less pronounced anomalies. Factor analysis enables recognition of different influences and, thus, was the best in differentiating natural and anthropogenic influences in the investigated estuary. The analysis of our data demonstrated that the choice of the dataset is also important in obtaining reliable results for statistical analysis. The presence of a few high values in the dataset may mask some natural geochemical anomalies. The analysis of two datasets, done in this work, was useful in recognizing more clearly natural and anthropogenic influences in the less polluted part of the investigated area.

In the majority of sediments sampled in the Krka river estuary, only lithogenic and marine influences according to factor analysis are notable. Spatial distribution of measured toxic elements indicates that the Krka river estuary is mostly unpolluted with some restricted, locally polluted areas. However, in these areas, severe pollution with some eco-toxic elements was noted, and the significance of this pollution should be further investigated. Fortunately, a further pollution of this, otherwise clean estuary, is not expected as the main pollution sources existing in the area are now rigorously regulated.

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REFERENCES

- [1] Raspor, B. (2004) Elements and elemental compounds in waters and the aquatic food chain. In: Merian, E., Anke, M., Inhat, M. and Stoeppler, M. (Eds.) Elements and their compounds in the environment. Occurrence, analysis and biological relevance. Wiley VCH, 127-148.
- [2] Forstner, U. and Salmons, W. (2004) Elements and compounds in sediments In: Merian, E., Anke, M., Inhat, M. and Stoeppler, M. (Eds.) Elements and their compounds in the environment. Occurrence, analysis and biological relevance. Wiley VCH, 149-162.
- [3] Bogner, D., Ujević, I., Zvonarić, T., Barić, A. (2004) Distribution of selected trace metals in coastal surface sediments from the middle and south Adriatic Sea. *Fresenius Environmental Bulletin* 13, 11b, 1281-1287.
- [4] Martinčić, D., Kwokal, Ž. and Branica, M. (1990) Distribution of zinc, lead, cadmium and copper between different size fraction of sediments. II The Krka River Estuary and the Kornati islands (Central Adriatic Sea). *Science of Total Environment*, 95, 217-225
- [5] Prohić, E. and Kniewald, G. (1987) Heavy metal distribution in recent sediments of the Krka River Estuary - an example of sequential extraction analysis. *Marine Chemistry*, 22, 279-297.
- [6] Šurića, B. and Branica, M. (1995) Distribution of Cd, Pb, Cu and Zn in carbonate sediments from the Krka River Estuary obtained by sequential extraction. *Science of Total Environment*, 170, 101-118.
- [7] Bogner, D., Ujević, I. and Barić, A. (2005) Trace metals (Cd, Pb, Cu, Zn and Cr) distribution in sediment along east coast of the Adriatic Sea (Croatia). *Fresenius Environmental Bulletin* 14, 1, 50-58.
- [8] Oreščanin, V., Barišić, D., Mikelić, L., Lovrenčić, I., Maćefat Rožmarić, M. and Lulić, S. (2006) Impact of past industrial activity on the content of heavy metals in the soil/grass system in Šibenik (Croatia). 5th European Meeting on Chemical Industry and Environment, EMChIE 2006, 887-894.
- [9] Bilinski, H., Kwokal, Ž. and Branica, M. (1996) Formation of some manganese minerals from ferromanganese factory waste disposed in the Krka River Estuary. *Water Research* 30 (3), 495-500.
- [10] Oreščanin, V., Barišić, D., Mikelić, L., Lovrenčić, I., Rubčić, M., Rozmarić-Maćefat, M. and Lulić, S. (2004) Environmental Contamination Assessment of the Surroundings of the ex-Šibenik's Ferro-manganese Smelter, Croatia. *Journal of Environmental Science and Health, Part A-Toxic/Hazardous Substances & Environmental Engineering*, A39, 9, 2493-2506
- [11] Bilinski, H., Kwokal, Ž. & Branica, M. (1992) Processes affecting the fate of mercury in the Krka River Estuary. *Water Research* 26 (9), 1243-1253.
- [12] Mikac N., Roje, V., Cukrov, N. and Foucher, D. (2006) Mercury in aquatic sediments and soils from Croatia. *Arhiv za higijenu rada i toksikologiju*, 57 (3), 325-332.
- [13] Cuculić V, Cukrov N, Barišić D, Mlakar M. (2006) Uranium in sediments, mussels (*Mytilus* sp.) and seawater of the Krka river estuary. *Journal of Environmental Radioactivity* 85, 59-70.
- [14] Cukrov, N., Cuculić, V., Barišić, D., Mlakar, M. (2006) Spatial distribution of uranium and radium in the sediments, mussels (*Mytilus* sp.) and sea water in port of Šibenik. *Promet - Traffic - Traffico* 18 (2), 89-93.
- [15] Juračić M. and Prohić E. (1991) Mineralogy, Sources of particles, and sedimentation in the Krka River estuary (Croatia). *Geološki Vjesnik* 44, 195-200.
- [16] Cukrov N., Barišić D. and Branica M. (2004) Content of the ^{40}K , ^{232}Th , ^{226}Ra , ^{238}U and ^{137}Cs in the recent sediments of the Krka river estuary. XXXVIIIth CIESM. Congress Proc., 37, 20.
- [17] Prohić E. and Juračić M. (1989) Heavy Metals in sediment – Problems Concerning Determination of Anthropogenic Influence. Study in the Krka River Estuary, Eastern Adriatic. *Environmental Geology and Water Sciences*, 13 (2), 145-151.
- [18] Lojen, S., Dolenec, T., Vokal, B., Cukrov, N., Mihelčić, G. and Papesch, W. (2004) Isotopic characteristics of tufa in the river Krka, Croatia: Relation to climatic conditions. *Sedimentology*, 51 (2) 361-375.
- [19] Cukrov, N. and Barišić, D. (2006) Spatial Distribution of ^{40}K and ^{232}Th in Recent Sediments of the Krka River Estuary. *Croatica chemica acta*, 79 (1), 115-118.
- [20] Cukrov, N., Barišić, D. & Juračić, M. (2007) Calculated sedimentation rate in the Krka River estuary using vertical distribution of ^{137}Cs . 38th CIESM Congress Proc. 81.
- [21] Förstner U. (2004) Traceability of sediment analysis, TRAC–Trends in Analytical Chemistry 23, 217-236.
- [22] Frančišković-Bilinski, S. (2008) Detection of geochemical anomalies in stream sediments of the Upper Sava River drainage basin (Slovenia, Croatia). *Fresenius Environmental Bulletin*. 17(2), 188-196.
- [23] Tukey, J.W. (1977) Exploratory data analysis. Reading: Addison-Wesley.
- [24] Reimann, C., Filzmoser, P. and Garrett, R.G. (2005) Background and threshold: critical comparison of methods of determination. *Science of the Total Environment* 346, 1-16.
- [25] Davis, J.C. (1986) Statistics and data analysis in geology, Second Edition. John Wiley & Sons, New York, Chichester, Brisbane, Toronto, Singapore
- [26] Halamić, J., Peh, Z., Bukovec, D., Miko, S., and Galović, L. (2001) A factor model of the relationship between stream sediment geochemistry and adjacent drainage basin lithology, Medvednica Mt., Croatia. *Geologia Croatica*, 54 (1), 37-51
- [27] Morrison, D.F. (1967) Multivariate statistical methods. Mc Graw-Hill, New York
- [28] Nathan, Y., Soudry, D., Levy, Y., Shitrit, D. and Dorfman, E. (1997) Geochemistry of cadmium in the Negev phosphorites. *Chemical Geology*, 142, 87-107.

- [29] Baïoumy, H.M. (2005): Preliminary data on cadmium and arsenic geochemistry for some phosphorites in Egypt. *Journal of African Earth Sciences*, 41, 266-274.
- [30] Zanin Yu. N. and Zamirailova, A. G. (2007) Trace Elements in Supergene Phosphorites. *Geochemistry International*, 45 (8), 829-841.
- [31] Mikac, N and Branica, M. (1992) Distribution of total and organic lead in Šibenik region (Eastern Adriatic). *Chem. Speciat. Bioavail.*, 4, 109-115.
- [32] NOAA (1999). National Oceanic and Atmospheric Administration, Sediment Quality Guidelines developed for the National Status and Trends Program (<http://archive.orr.noaa.gov/cpr/sediment/SPQ.pdf>)
- [33] ANZECC & ARMCANZ (2000) Australian and New Zealand Guidelines for Fresh and Marine Water Quality. Volume 1, Section 3.5: Sediment Quality Guidelines. Australian and New Zealand Environment and Conservation Council, and Agriculture and Resource Management Council of Australia and New Zealand. Canberra, Australia
- [34] Müller, G. (1979). Schwermetalle in den Sedimenten des Rheines – Veränderungen seit 1971. *Umschau* 79, 778-785.
- [35] Turekian, K.K. and Wedepohl, K.H. (1961). Distribution of the elements in some major units of the earth's crust. *Geological Society of America Bulletin* 72, 175-192.
- [36] New York State Department of Environmental Conservation (1999). Technical Guide for Screening Contaminated Sediments (http://www.dec.ny.gov/docs/wildlife_pdf/seddoc.pdf)

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